Joseph Li

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Education

University of Southern California — Los Angeles, CA
M.S. Computer Science Engineering, Artificial Intelligence; GPA: 3.50/4.00 — *Sep 2024 – May 2026*

University of Michigan — Ann Arbor, MI
B.S.E. Data Science Engineering; GPA: 3.68/4.00 — *Sep 2020 – May 2024*

Experience

AI Engineer — Aug 2025 – Dec 2025
Yoyo Markets — *New York, NY (Remote)*

- Led a 3-person team building an AI pipeline that mines real-time social signals to generate prediction markets, producing 50+/week (5x over the prior manual process)
- Designed a multi-agent ELO ranking system evaluating 200+ candidate markets per cycle, improving market quality scores by 30% and removing the need for manual curation entirely
- Wrote a custom async Python scraper targeting 150+ high-signal KOL accounts on Twitter, expanding real-time signal coverage by 3x beyond public trend aggregators
- Shipped a Next.js internal dashboard on Vercel used daily by the founding team to review, approve, and publish markets, cutting end-to-end publishing time from 30 minutes to under 5

Machine Learning Engineering Intern — Jun 2025 – Aug 2025
Pure Global — *Jersey City, NJ (Remote)*

- Built a RAG-powered agent that classifies medical devices into FDA regulatory pathways (510(k), PMA, De Novo), cutting manual compliance research time by over 90%
- Set up the full pipeline with n8n for orchestration and LangChain for retrieval over 150,000+ FDA documents, hitting 88% classification accuracy on internal benchmarks
- Built a frontend interface so compliance teams could query the system and review results directly, replacing a workflow that previously took days of manual research

Projects

Autonomous Research Agent | *Python, LangGraph, FAISS, FastAPI* | github.com/josephl894 — 2025

- Built an agentic research tool that decomposes a question into sub-queries, searches arXiv and web sources in parallel, and synthesizes findings into a structured report with citations
- Implemented tool-calling orchestration with LangGraph, FAISS vector store for source deduplication, and streaming output for real-time report generation

Privacy-Preserving Clinical Text Obfuscation | *Python, NLP, Differential Privacy* | *UMich SOCR Lab* 2023 – 2024

- Built an NLP obfuscation algorithm for unstructured clinical text using named entity recognition and differential privacy, enabling safe use of 10,000+ patient records in published research
- Benchmarked across 3 NLP pipelines and achieved 94% entity redaction accuracy while keeping downstream task performance within 3% of the unmasked baseline

Technical Skills

Languages: Python, C++, TypeScript, SQL
ML & AI: PyTorch, LangChain, LangGraph, scikit-learn, RAG, Multi-Agent Systems, NLP, Fine-Tuning
Frameworks & Tools: Next.js, FastAPI, Flask, React, Docker, Git, n8n, Vercel, pytest, CI/CD
Data & Infrastructure: AWS (S3, Lambda), PostgreSQL, FAISS, Vector Databases, REST APIs